

Personal Research Ontologies for Personal Open Science

Jonas Oppenlaender
Centre for Applied Computing
University of Oulu
Oulu, Finland
jonas.oppenlaender@oulu.fi

Abstract

As researchers conduct their work, they incrementally construct a *personal research ontology*: a rich, personal understanding of their field and its phenomena. Yet most of this knowledge remains internalized, with only fragments surfacing in publications. This position paper argues that research is fundamentally an act of personal ontology construction and proposes *Personal Open Science* (POS), a lightweight method for externalizing a researcher's personal research ontology for downstream use. The method requires no machine learning or language models, and it centers on the researcher rather than on the publication, the traditional focal unit of Open Science. Building on daily reading habits and a minimal annotation scheme, the approach adds little overhead while producing a structured, explorable representation of a researcher's interests, influences, and expertise. I demonstrate the method through my own practice of annotating over 45,000 documents across a decade of research. I discuss how externalized personal research ontologies could serve multiple downstream purposes, such as the discovery of collaborators and self-reflection, by revealing the gap between a researcher's understanding and the shared ontology of their field. When many researchers externalize their ontologies, the resulting network of overlapping and complementary knowledge structures becomes a resource for collective intelligence, making visible the distributed expertise of a research community and enabling new forms of intellectual coordination across disciplinary boundaries.

CCS Concepts

• **Computing methodologies** → **Ontology engineering**; • **Human-centered computing** → *Collaborative and social computing theory, concepts and paradigms*; • **Applied computing** → Document management and text processing.

Keywords

Open Science, knowledge externalization, collective intelligence, scholarly communication

ACM Reference Format:

Jonas Oppenlaender. 2026. Personal Research Ontologies for Personal Open Science. In *Proceedings of the ACM Collective Intelligence Conference (CI '26)*, September 27–30, 2026, Alexandria, VA, USA. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3834581.3838630>

1 Introduction

An ontology is an explicit specification of a conceptualization.

—Tom Gruber, 1993

This position paper starts from a simple observation: research is fundamentally an act of personal ontology construction. The term *ontology* originates in philosophy, where it denotes the study of what exists. In computer science, the term was adopted to mean an explicit specification of a conceptualization [21, 22]. Every researcher constructs a conceptualization through daily practice, by reading scientific articles, sorting findings into categories, observing connections and patterns, and identifying open research questions. I call this evolving conceptualization a **personal research ontology**. The personal research ontology is the conceptual structure through which a researcher makes sense of a field: the topics they track, the connections they draw among them, and the relative significance they attach to different ideas and findings. Most of this personal ontology remains implicit, locked inside the researcher's head. This opacity differs from tacit knowledge (i.e., the non-codified know-how acquired through informal socialization) in organizational theory [25]. A personal research ontology is personal knowledge [42]. This knowledge is, in principle, codifiable: it is the conceptual structure that is shaped by how a researcher engages in sense-making with respect to the scientific literature and their discipline, how a researcher selects problems, interprets evidence, and navigates their field. This structure is meaningful, but in practice, it is not directly observable. In the qualitative research tradition, the accumulation of internalized experiences is valued as personal and professional growth [13, 30]. But internalized knowledge is fragile: we tend to forget, or we simply move on to other topics. And the personal experience cannot be searched, shared, or built upon by others.

Open Science initiatives have made progress in making research outputs visible and accessible [11, 14, 15]. Executable notebooks [48] and research objects [6, 45] were once heralded as successors of the scientific paper. Yet, while preprints, datasets, and code repositories are now openly available, current publishing remains centered on the publication as the primary unit of scientific communication. A researcher's Google Scholar profile, for instance, lists their published works but reveals nothing about the thousands of papers they intellectually engaged with in the process of 'generating' these works, the topics they explored but never published on, their latest (unpublished) research interests, or how they organize their understanding of the field. Thus, a researcher's publications provide only a narrow and often outdated window into the researcher's personal research ontology.



This work is licensed under a Creative Commons Attribution 4.0 International License. <https://creativecommons.org/licenses/by/4.0/>

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2895-2/26/09

<https://doi.org/10.1145/3834581.3838630>

Meanwhile, the landscape of scientific production is shifting. With generative artificial intelligence, the marginal cost of producing paper-shaped research output is collapsing to zero [8, 27]. Generative AI is diminishing the reliability of publication-based indicators. Furthermore, traditional measures of research impact, such as the h-index, cannot distinguish deep expertise from surface-level familiarity, and contemporary co-authorship practices further dilute the signal [1, 29, 35]. New ways of making research expertise visible are needed.

This paper makes three contributions. First, it introduces the concept of the **personal research ontology** and argues that constructing such an ontology is the core intellectual work of researchers. Second, the paper proposes **Personal Open Science (POS)**, a vision and a lightweight method for externalizing the personal research ontology. The method places the researcher, rather than the publication, at the center of Open Science. I demonstrate the method through my own practice of annotating over 45,000 documents across a decade of research. Third, the paper discusses what becomes possible when many researchers externalize their ontologies: a network of overlapping knowledge structures that makes the distributed expertise of a research community visible and supports new forms of collective intelligence.

2 Background

2.1 Open Science and the Primacy of the Publication

The central tenet of Open Science is to make scientific research and its dissemination accessible to all levels of society [33]. The movement encompasses far more than free access to scientific articles. Fecher and Friesike identify five schools of thought within Open Science [17], ranging from opening the research process itself to measuring impact in new ways. Initiatives such as the European Open Science Cloud aim to provide federated environments where researchers can publish, find, and reuse data, tools, and services [16], and journals increasingly adopt guidelines that promote transparency and reproducibility [39, 53].

In practice, Open Science is frequently conflated with Open Access. This reduces a multi-faceted concept to a single binary dimension: whether to publish in an open access format or not. Researchers who follow the movement typically upload preprints to repositories [54], publish datasets and code, and write open peer reviews. These artifacts orbit the same focal unit: the scientific publication, a practice dating back to the emergence of the academic journal in the 17th century. A gap persists between the grand vision of Open Science and what researchers do [38].

This publication-centrism reflects a broader tension between object-centric and people-centric views of knowledge work [2, 19], and it has a practical cost: because journal articles may take two to three years to be accepted, a researcher's profile often reflects neither current work nor latest interests. The researcher receives insufficient attention in Open Science, even though academia has a culture of professional openness rare in other sectors.

2.2 Making Hidden Work Visible

Some researchers have tried to make the hidden work in science visible. One approach is the “CV of failures” (or “shadow CV”),

which lists a researcher's rejected papers, failed funding bids, and unsuccessful fellowship applications. The practice opposes the bias of only reporting successes [49], but has not found widespread adoption. To my knowledge, only four researchers in the fields of Computing and Human-Computer Interaction have published such documents: Jessica Vitak,¹ Johannes Haushofer,² Melanie Stefan,³ and Veronika Cheplygina.⁴ Maintaining such a list requires effort without offering much in return, which is why this trend was short-lived. The lesson matters for any proposal in this space: a practice of making hidden work visible must be nearly free to sustain yet add value to the person who maintains the practice.

More systematic efforts have come from the publishing world. “Enhanced publications” were a trending topic about 15 years ago promising to link articles to their underlying data and other research objects in digital repositories [58]. “Publishing 3.0” [55] and interactive, computational papers [48] reimagined the article itself. But these efforts found traction with only a few publishers. Because these efforts depend on publishers, researchers have contributed little to their uptake. More importantly, they keep the publication as their focal point. Research information systems, such as VIVO [10] and Harvard Catalyst [23], move toward researcher-centered infrastructure by aggregating profiles for networking [3], but they populate these profiles primarily from the publication record. What a researcher explores, values, and knows remains invisible in all of these approaches.

The closest predecessor to the approach proposed in this paper is Luhmann's *Zettelkasten*, a slip-box system in which the sociologist externalized his personal knowledge structure as a hierarchy of handwritten, cross-referenced notes [46]. His practice demonstrated that making one's conceptual organization explicit can produce new insights and connections, not merely archive existing ones. Yet even Luhmann's system required writing, numbering, and cross-referencing each note by hand, an investment that few researchers sustain at scale.

A parallel practice has developed largely outside academia. Since 2008, the Quantified Self movement [51] has brought together individuals who use empirical methods to answer questions about their own lives, a practice its proponents call *personal science* [57]. These practitioners study themselves rather than a scientific field, yet they face the problem this paper starts from: each accumulates situated knowledge that is valuable to them and largely illegible to anyone else. When the community tried to pool that knowledge, the difficulty surfaced. Kloppenborg et al. co-designed the Personal Science Wiki with practitioners and found in a user study that people organized the same material in different ways, and that one person often placed a single item in several categories at once [28]. The design settled on multiple paths through the material rather than one hierarchy. Making personal knowledge collective is, in part, a problem of classification.

¹<https://web.archive.org/web/20240715184947/https://jessicavitak.com/2016/04/30/embracing-failure-heres-my-cv-of-academic-rejections/>

²https://web.archive.org/web/20240324054716/https://crlte.engin.umich.edu/wp-content/uploads/sites/5/2020/05/Johannes_Haushofer_CV_of_Failures.pdf

³<https://www.nature.com/articles/nj7322-467a>

⁴<https://web.archive.org/web/20250908200410/https://veronikach.com/failure/shadow-cv/>

2.3 Folksonomies and Social Tagging

The idea that individuals can externalize how they classify what they encounter online is not new. In the mid-2000s, a wave of Web 2.0 social bookmarking and social tagging services [31, 40] let users label web pages, photographs, and references with free-form keywords. Services such as del.icio.us and Flickr popularized the practice, and the bibliographic services CiteULike, BibSonomy, and Mendeley brought it to scientific literature, letting researchers tag and share the papers they read [24]. The aggregate of these user-assigned tags was called a *folksonomy*, a classification that emerges bottom-up from many individual acts rather than being imposed top-down by experts [20, 32]. A lively debate followed about whether such bottom-up tagging could substitute for, or only complement, the formal ontologies of knowledge engineering [36, 47].

The fate of this wave is instructive. Some of these services did not survive. CiteULike shut down in 2019, and social bookmarking for science never reached the scale its proponents imagined. The signal that Mendeley readership counts provided was studied as an early indicator in the science of science [37, 43], yet the practice of tagging imposed an overhead that most researchers eventually abandoned. The reading behavior of scientists carries a useful signal, but capturing it has proven difficult to sustain, a lesson this paper returns to in Section 5.6. The underlying problem was one of incentives: tagging was extra work performed primarily for the benefit of the collective, not for the individual researcher. When the collective benefit failed to materialize at scale, the individual effort lost its justification.

2.4 Ontologies and Collective Intelligence

In knowledge engineering [50] and the field of the Semantic Web [7, 52], ontologies were developed as formal, explicit specifications of a shared conceptualization in a domain of interest [50], designed so that knowledge could be exchanged between systems and reused [21]. A scientific field can be viewed in the same way. The field's taxonomy of topics, its canonical references, and its sense of which problems matter form a shared ontology that members of the field carry, imperfectly and partially, in the public record of publications and in their heads.

This view connects directly to collective intelligence. Groups perform well when they develop a shared understanding of who knows what – a phenomenon called transactive memory [56]. In science, this capability is weak. Scientists self-organize into teams, but team formation comes at a considerable cost. A field experiment at Harvard Medical School showed that even small search costs affect who collaborates with whom [9]. The information needed to identify good collaborators, namely researchers with deep knowledge of specific topics, exists, but it is distributed across thousands of heads and only weakly signaled by publication records. Making personal ontologies explicit is, in this framing, a way of building a transactive memory system for science.

3 From Implicit Knowledge to Personal Open Science

3.1 Personal Research Ontologies

Constructing a personal research ontology is not incidental to research. It is what much of research is about. Reading a paper means deciding where it fits in a broader knowledge structure. Conducting a literature review means restructuring a region of the ontology. Identifying a research gap means noticing a place where the ontology is thin.

Three properties matter for this paper. First, the ontology is *personal*. It is shaped by the researcher's interests, training, personality, curiosity, intellectual capacity, and environment (in the sense of what Rhodes called "press" in his four-P model of creativity [44]). Two researchers in the same field will organize their ontologies differently, and those differences carry meaning. Second, the ontology is *evolving*. It changes with every paper read and every project completed. At any moment it reflects the researcher's current understanding, unlike the publication record, which trails years behind. Third, the ontology is *broadier than the publication record*. It covers everything the researcher has engaged with, including topics they explored but never published on. The published record samples only a thin slice of it.

Each personal ontology relates to the shared ontology of the researcher's field. A personal ontology is a partial and idiosyncratic projection of this shared structure. The overlap between the two indicates how well the researcher has internalized the field. The divergence indicates either a gap in learning or, sometimes, an original perspective on topics the field has not yet charted. This relation grounds the use of personal ontologies as building blocks of collective intelligence (see Section 5) and for self-assessment (Section 5.3).

The problem is that this personal research ontology is implicit. The next section proposes a method that externalizes a useful approximation as a by-product of daily work. The externalization need not capture the full ontology to be useful. Even a partial, structured trace of the researcher's engagement with the literature carries information that the publication record does not.

3.2 Personal Open Science: A Vision and a Method

Personal Open Science (POS) is both a vision and a lightweight method. The POS vision holds that Open Science should extend beyond published artifacts. Traditional Open Science makes publications and their associated artifacts openly available. Personal Open Science complements this by making visible the broader knowledge structure that a researcher builds through daily practice. The POS method is one instantiation of this vision: a deliberately lightweight approach combining a long-term reading habit with a minimal annotation scheme. Everything else, such as the visualizations and the question-answering interface described in Section 4, is a downstream use of the annotated corpus. This section describes the method as I have practiced it since 2016. It is an existence proof, not the only possible instantiation. Because the method centers on reading and curation, it captures primarily one dimension of

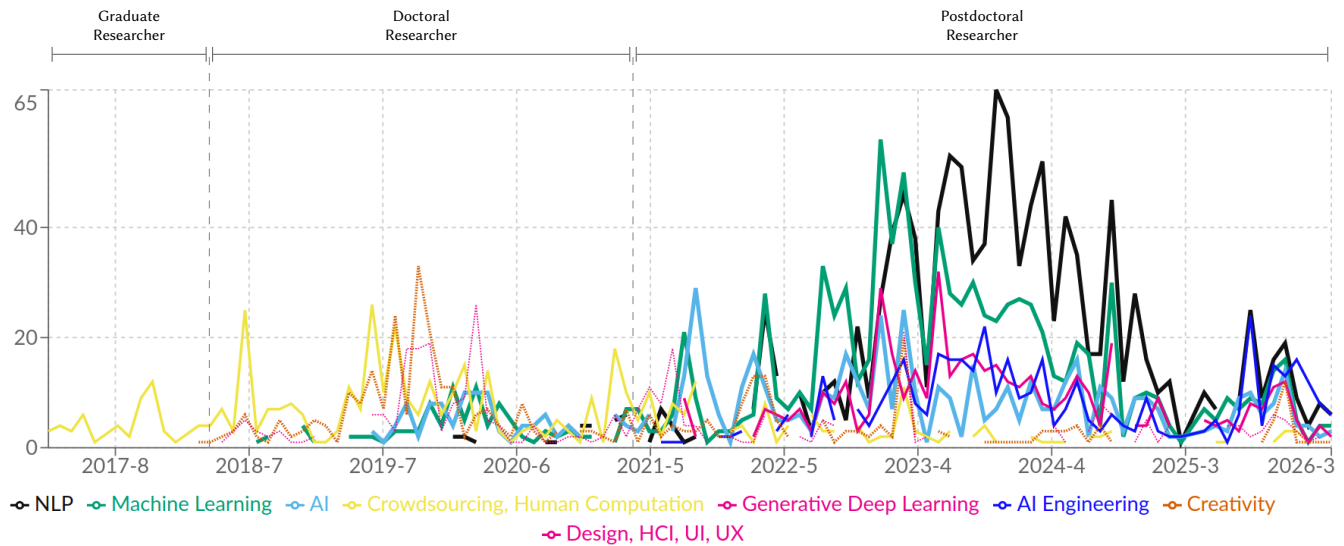


Figure 1: Reading activity over time across selected top-level topics, derived from timestamps of documents rated as relevant or important (“!” prefix or higher) in the annotated collection. Shifts in attention, such as the rise of generative AI topics from 2022 onward, become visible without any manual tracking. The brackets above the plot mark three career stages: graduate researcher until April 2018, doctoral researcher until April 2021, and postdoctoral researcher thereafter.

the personal research ontology. Future extensions could externalize other dimensions, such as research activities, methodological decisions, and collaborative interactions, to approximate the full personal research ontology more closely.

3.2.1 Design goals. The method is designed around a simple set of goals:

- (D1) Integration into the daily workflow,
- (D2) Low effort and friction,
- (D3) Value added, and
- (D4) Support for exploration and self-reflection.

The first two goals follow from the lesson of Section 2.2, and together the design goals respond to the incentive failure of folksonomies (Section 2.3): where social tagging asked researchers to do extra work for a collective payoff that might never arrive, POS embeds the annotation into the researcher’s daily work and delivers individual value from day one (D3; see Section 4.1). A method for Personal Open Science becomes more useful the longer it is sustained. The accumulated data must provide more value than the individual pieces alone (D3), and the structure the method produces should allow the researcher to explore and reflect on it (D4).

3.2.2 Annotation scheme and consistent habits. The core of the method is a habit I developed during my Master’s degree and solidified during my first graduate research position. Some researchers manage their literature by writing detailed notes on each paper they read, for instance in notebooks, spreadsheets, or annotated bibliographies. Reference managers such as Zotero organize bibliographic metadata, and personal knowledge management tools such as Obsidian, Roam, and Notion offer environments for linking notes and ideas in the tradition of Luhmann’s *Zettelkasten*. These

practices produce valuable records of engagement with the literature, but they impose a significant recurring cost on daily work. A few minutes of note-taking or curation per paper accumulate to hours each week. A practice intended to run for years must be lighter than that (D1, D2). The method proposed here is not another tool. It requires no separate applications, only the file system every researcher already uses. The method supports continuous foraging of information with minimal overhead.

The method’s premise is that anything I read may become relevant later. By consistently applying the method, I configure my informational environment for future serendipitous discovery. Thus, the first element of the method is a habit related to storage: I save everything I read as PDF documents on my computer. Whether it is a scientific paper, a Wikipedia article, a blog post, or a BBC news article, every document enters the same archive. If needed, optical character recognition (OCR) is directly applied in my PDF reader before saving the document. The archived PDFs serve as the raw material for all downstream uses described in Section 4.

The second element is a lightweight annotation. When saving a document, after reading or skimming it, I assign a coarse importance score encoded in the filename (see Table 1).

I deliberately do not distinguish between relevance and importance. A literature review on a topic outside my current interests is still important and may become relevant in future research. I intentionally do not keep notes or write summaries of articles. I started this habit in 2016 with the vision of one day querying the full text of the documents with language models, which would make manually written summaries redundant.

The method’s third element is a hierarchical folder structure. The top level distinguishes three areas: “Literature” (the most relevant material, accessed daily), “Academic Working” (literature

Table 1: The annotation scheme. The importance score is encoded as a prefix in the filename, which keeps the annotation visible in every tool that lists files. The remainder of the filename is left as the publisher assigned it.

Prefix	Meaning	Example filename
(none)	not yet classified	2108.13161.pdf
-	not relevant or important	-1810141115.full.pdf
!	relevant or important	!1478744.pdf
!!	very relevant or important	!!782780.pdf
!!!	extremely relevant or important	!!!p1245-yu.pdf

about science itself), and “Miscellaneous” (less relevant material). Below that, folders represent topics and may nest to arbitrary depth. For instance, the folder “Creativity” contains a folder “Creativity Support Tools,” which in turn contains other folders. If there are synonyms for a topic, a comma in the folder name separates them. Three reserved folder names recur throughout the folder hierarchy and encode consistent semantics. “Applications” collects articles on how a method or technology is applied. “Issues” collects open problems and challenges of a topic. “General, Theory” collects general information on fundamental concepts of a topic.

This is the entire annotation scheme: an importance prefix and a place in a topic hierarchy, assigned at the moment of saving a file. The marginal effort per document is a few seconds (D1, D2). After a decade of practice, the structure comprises 45,444 PDF documents in 28,075 folders, with a maximum folder depth of 16. The folder hierarchy, the importance prefixes, and the file timestamps together form an externalized approximation of one part of my personal research ontology: what topics I distinguish, how I relate them, which documents I attach to them, how much I value each document, and when I engaged with it.

Formally, the folder hierarchy is an annotated taxonomy, not an ontology in Gruber’s sense. The knowledge structure is deliberately designed as a simple ontology [34]. The personal research ontology, as defined in Section 3.1, should be considered as the complete conceptual structure in the researcher’s head: the topics, the connections drawn across them, the methodological preferences, the sense of which problems matter. The folder hierarchy externalizes one facet of this structure: the topics the researcher distinguishes, their nesting, and a coarse importance judgment on each document. Other facets remain implicit, and the gap between the full ontology and its externalized approximation is where future work lies. For the uses proposed in this paper, a lightweight structure that a researcher maintains is more useful than a formal ontology that nobody builds.

4 Downstream Uses: What the Externalized Ontology Enables for the Individual

The method described above produces a document corpus organized into a topic hierarchy, rated by importance. This section describes what I have built on top of this corpus, namely a public web interface that visualizes the ontology and a question-answering interface. These are two downstream uses, not part of the POS method. Other researchers could build different applications on the same kind

of corpus. The section outlines the value a decade of practice has shown at the individual level. I describe the individual benefits before any collective ones, because a practice must benefit the individual researcher before any collective benefit can materialize (D3). Section 5 broadens the perspective to collective intelligence.

4.1 An External Memory and Object for Self-Reflection

The hierarchical document storage functions as an external workbench onto which I offload memory. I often cannot recall what I read a few days ago, but I can find it in the list of recently read documents. A simple shell command can surface recently read documents by timestamp. When I need the most relevant literature on a specific topic, I navigate the folder hierarchy and let the importance prefixes surface the documents that matter. The full text remains searchable as a fallback. The ontology converts a vague sense of “I have read something about this” into a retrievable artifact.

Because the ontology is externalized, I can examine it as a whole. The folder structure shows me where my reading has been deep and where it has been shallow. The reading timeline shows how my interests have shifted (Figure 1). Researchers routinely describe their interests in abstract terms. An externalized ontology lets me check these descriptions against evidence about my behavior (D4).

4.2 Publishing and Visualizing the Ontology

The annotated corpus becomes Personal Open Science when parts of it are published. Periodically, I run a script that traverses the folder structure, extracts metadata, and regenerates several interactive views on the ontology. Titles and authors are parsed from the PDFs using Ai2’s Science Parse, which succeeds for about two thirds of the documents. Where parsing fails, my manual annotation of the title is used, yielding 100% paper title coverage. The last-modified date of each file indicates when I last read or annotated it. The resulting data feeds several views on my personal website (Figure 1, Figure 2, and Figure 3).

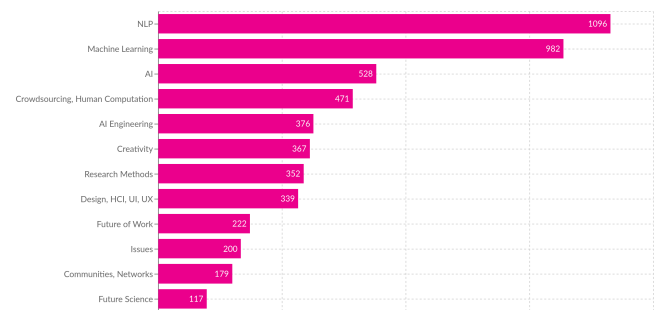


Figure 2: Screenshot of an interactive visualization of the largest first-level topics of the folder hierarchy, ranked by the number of documents rated as relevant or important (“!” prefix or higher). A view of this kind replaces the keyword list of research interests found on most academic websites with counted evidence of sustained attention.

What am I reading? The website lists the last hundred documents I most recently read. Assembling this list adds little overhead in my